Supplementary Material of "Temporal Smoothness Doubly Robust Learning for Debiased Knowledge Tracing"

The supplementary material is organized as follows: Appendix A presents the sources of the public datasets and their statistical information; Appendix B describes the details of how our simulated data were generated, including the statistical properties of the data and tables showing the performance of different models on them; Appendix C introduces the baseline models we employed; Appendix D further supplements the ablation study and computational overhead analysis in response to the reviewers' comments. It summarizes the contribution of each component in the proposed DR module and reports the additional training cost introduced by DR; Appendix E outlines our main theoretical results to facilitate understanding of the proposed method; and Appendix F provides the proofs corresponding to the theoretical results in Appendix E.

Appendix A Details of Benchmark Dataset

We evaluate the performance of TSDR on 9 public datasets:

- **Spanish** [Lindsey *et al.*, 2014]: The spanish dataset is from middle-school students practicing Spanish exercises, including translations and applications of basic skills like verb conjugation, over a 15-week semester.
- **ASSISTments17** [Patikorn *et al.*, 2018]: The ASSISTments17 dataset is derived from the 2017 Longitudinal Data Mining Competition, containing longitudinal student interaction data collected from the ASSISTments online tutoring system.
- **Slepemapy** [Papoušek *et al.*, 2016]: The slepemapy dataset originates from slepemapy.cz, an online platform dedicated to the adaptive practice of geography facts.
- **Algebra05** [Stamper *et al.*, 2010]: The algebra05 dataset comes from the KDD Cup 2010 EDM Challenge, containing detailed step-level student responses to algebra questions from the Carnegie Learning system.
- **Prob, Linux, Database, Comp (PTADisc)** [Hu *et al.*, 2023]: The prob, comp, linux, database datasets are collected from the Programming Teaching Assistant platform, specifically from course exercises in Probability and Statistics, Computational Thinking, Linux System, and Database Technology and Application.
- **EdNet** [Choi *et al.*, 2020]: The ednet dataset, collected by the multi-platform AI tutoring service Santa, stands as the largest publicly released interactive educational system dataset to date.

Dataset	# Users	# Questions	# concepts	# Interactions
Spanish	182	409	221	578,726
ASSIST17	1,708	3,162	411	934,638
Slepemapy	5,000	2,723	1,391	625,523
Algebra05	571	173,113	271	607,014
Prob	512	1,054	247	42,869
Linux	4,375	2,672	281	365,027
Database	5,488	3,388	291	990,468
Comp	5,000	7,460	445	668,927
EdNet	5,000	12,002	1,769	789,812

Table 1: Statistics of the 9 datasets

Table 1 presents the statistics of the processed datasets, excluding student sequences with fewer than 5 interactions.

Appendix B Details of the Simulation Study

This section details the data generation process for our simulation study. The base dataset was generated for 1,000 students and 200 questions, with the questions covering 20 distinct knowledge components. The experimental results on this synthetic dataset are presented in Table 3. The process incorporates four key characteristics:

1. **Long-tail Distribution**: To mimic the real-world scenario where some knowledge points are practiced far more often than others, the frequency of items (questions) was modeled to follow a Zipfian distribution with a shape parameter of $\alpha = 0.8$.
2. **Dynamic Learning**: A student's mastery of a knowledge component is not static. We simulated dynamic learning where a student's mastery level evolves over time. Upon answering a question correctly, their mastery of the associated knowledge components increases with a learning rate of 0.55.
3. **Response Noise**: Student responses are not always a perfect reflection of their true mastery. To account for this uncertainty, we introduced a guessing probability of 0.1 (the probability of a non-proficient student answering correctly) and a slipping probability of 0.05 (the probability of a proficient student answering incorrectly).

4. **Controllable MNAR Mechanism**: This is the core mechanism for simulating selection bias. Interactions are selectively marked as "skipped" (missing) based on the ground-truth probability of success, denoted as p. This simulates students strategically avoiding questions they perceive as too difficult or too easy. The skipping probability is determined by the MNAR degree γ and the extremeness of p:
   - Too difficult ($p < 0.25$): The skipping probability is $\gamma \cdot (1 - p)$, meaning students are more likely to skip harder questions (lower p).
   - Too trivial ($p > 0.75$): The skipping probability is $\gamma \cdot p$, meaning students are more likely to skip easier questions (higher p).

   To create a spectrum of bias scenarios for our robustness evaluation, we generated six distinct datasets by setting the MNAR degree γ to values across the set $\{0, 0.2, 0.4, 0.6, 0.8, 0.999\}$. A γ of 0 corresponds to a dataset with no selection bias, while a γ of 0.999 represents a very strong selection bias. Dataset statistics are summarized in Table 2.

Dataset	Degree (γ)	# Observed	# Skipped	# Interactions
Synthetic-1	0.0	114482	0	48,967
Synthetic-2	0.2	116791	12922	48,476
Synthetic-3	0.4	114214	24669	48,578
Synthetic-4	0.6	114387	36055	48,091
Synthetic-5	0.8	114789	46627	47,826
Synthetic-6	0.999	114191	55770	47,645

Table 2: Statistics of the generated synthetic datasets under varying degrees of Missing Not At Random (MNAR).

Appendix C Details of Baseline Models

To evaluate the universality and effectiveness of our proposed TSDR framework, we selected 6 representative and state-of-the-art Knowledge Tracing models as backbones. These baselines include models built on Recurrent Neural Networks (RNNs) and attention mechanisms. The details of each baseline are as follows:

- **DKT** [Piech *et al.*, 2015]: DKT is the pioneering work that introduced deep learning to knowledge tracing. It utilizes LSTM networks to model the student's knowledge state as a hidden vector that evolves over time. DKT captures the sequential dependencies in the interaction history to predict future performance.
- **AKT** [Ghosh *et al.*, 2020]: AKT adapts the Transformer architecture for KT tasks. It introduces a context-aware monotonic attention mechanism to weigh the importance of past interactions relative to the current question. AKT significantly improves interpretability and performance by explicitly capturing the decay of memory over time using exponential decay kernels.
- **simpleKT** [Liu *et al.*, 2023]: simpleKT challenges the complexity of previous models by demonstrating that a simple but strong baseline can outperform complex architectures. It focuses on capturing question-specific variations using specific embedding strategies without heavy reliance on complex attention layers, highlighting the importance of input feature engineering.
- **FoLiBiKT** [Im *et al.*, 2023]: Addressing the limitation that standard attention mechanisms may not fully capture human memory characteristics, FoLiBiKT incorporates a forgetting-aware linear bias into the attention scores. It explicitly models the forgetting curve based on the time interval and the number of intervening interactions, making the attention mechanism more cognitively plausible.
- **SparseKT** [Huang *et al.*, 2023]: Recognizing that standard attention mechanisms often assign non-zero weights to irrelevant history (noise), sparseKT introduces a k-sparse attention mechanism. It filters out noisy or irrelevant historical interactions by retaining only the top-k most relevant attention scores, thereby enhancing the model's robustness and focusing on the most critical evidence for mastery tracing.
- **DisKT** [Zhou *et al.*, 2025]: disKT aims to solve the confounding issues in student performance prediction. It utilizes a disentangled representation learning framework to separate the student's intrinsic ability from the question's difficulty and other confounding factors. By explicitly modeling these factors separately, disKT achieves better generalization and interpretability, representing the state-of-the-art in handling cognitive biases in KT.

Appendix D Details of the Ablation Study and Computational Overhead

To address the reviewers' concerns about the contribution of each component and the additional computational cost, we further conduct an ablation study and report the wall-clock time and memory overhead of the proposed DR module. Specifically, we investigate two key questions: (1) whether joint learning brings consistent benefits, and (2) where temporal smoothness should be imposed to better support knowledge tracing. The results are summarized in Tables 4 and 5.

Table 4 presents the ablation results across six representative KT backbones. Compared with using DR alone, introducing joint learning generally improves the prediction performance, indicating that the imputation model provides useful auxiliary supervision for the KT backbone. Moreover, applying temporal smoothness to the imputation branch consistently achieves the best or near-best results across different backbones. In contrast, directly imposing temporal smoothness on the KT backbone often leads to inferior performance. This suggests that temporal smoothness is more suitable for regularizing the auxiliary imputation process, where it encourages stable and coherent latent hints, rather than directly constraining the KT backbone.

Overall, the full variant, i.e., DR with joint learning and temporal smoothness on the imputation branch, obtains the

Dataset	Metric	DKT	AKT	simpleKT	FoLiBiKT	SparseKT	DisKT
Synthetic (MNAR 0.999)	AUC ↑	0.6569 / **0.6653** (↑ 1.28%)	0.6784 / **0.6960** (↑ 2.59%)	0.6906 / **0.7073** (↑ 2.42%)	0.6783 / **0.6918** (↑ 1.99%)	0.6952 / **0.7015** (↑ 0.91%)	0.7058 / **0.7233** (↑ 2.48%)
	ACC ↑	0.7459 / **0.7489** (↑ 0.40%)	0.7368 / **0.7485** (↑ 1.59%)	0.7442 / **0.7505** (↑ 0.85%)	0.7392 / **0.7503** (↑ 1.50%)	0.7388 / **0.7516** (↑ 1.73%)	0.7428 / **0.7554** (↑ 1.70%)
	RMSE ↓	0.4217 / **0.4203** (↓ 0.33%)	0.4228 / **0.4146** (↓ 1.94%)	0.4184 / **0.4129** (↓ 1.31%)	0.4198 / **0.4150** (↓ 1.14%)	0.4189 / **0.4138** (↓ 1.22%)	0.4157 / **0.4113** (↓ 1.06%)
Synthetic (MNAR 0.8)	AUC	0.6549 / **0.6697** (↑ 2.26%)	0.6806 / **0.6959** (↑ 2.25%)	0.6962 / **0.7044** (↑ 1.18%)	0.6803 / **0.6941** (↑ 2.03%)	0.7025 / **0.7160** (↑ 1.92%)	0.7115 / **0.7290** (↑ 2.46%)
	ACC	0.7420 / **0.7448** (↑ 0.38%)	0.7405 / **0.7476** (↑ 0.96%)	0.7419 / **0.7478** (↑ 0.80%)	0.7379 / **0.7470** (↑ 1.23%)	0.7389 / **0.7487** (↑ 1.33%)	0.7463 / **0.7524** (↑ 0.82%)
	RMSE	0.4242 / **0.4219** (↓ 0.54%)	0.4205 / **0.4167** (↓ 0.90%)	**0.4171** / 0.4176 (↑ 0.12%)	0.4209 / **0.4185** (↓ 0.57%)	0.4171 / **0.4119** (↓ 1.25%)	0.4148 / **0.4137** (↓ 0.27%)
Synthetic (MNAR 0.6)	AUC	0.6633 / **0.6664** (↑ 0.47%)	0.6823 / **0.7046** (↑ 3.27%)	0.6981 / **0.7114** (↑ 1.91%)	0.6851 / **0.6908** (↑ 0.83%)	0.7093 / **0.7202** (↑ 1.54%)	0.7148 / **0.7285** (↑ 1.92%)
	ACC	0.7386 / **0.7430** (↑ 0.60%)	0.7303 / **0.7500** (↑ 2.70%)	0.7435 / **0.7497** (↑ 0.83%)	0.7297 / **0.7482** (↑ 2.54%)	0.7381 / **0.7528** (↑ 1.99%)	0.7448 / **0.7561** (↑ 1.52%)
	RMSE	0.4239 / **0.4233** (↓ 0.14%)	0.4258 / **0.4154** (↓ 2.44%)	0.4182 / **0.4135** (↓ 1.12%)	0.4236 / **0.4199** (↓ 0.87%)	0.4175 / **0.4114** (↓ 1.46%)	0.4154 / **0.4125** (↓ 0.70%)
Synthetic (MNAR 0.4)	AUC	0.6685 / **0.6741** (↑ 0.84%)	0.6890 / **0.7127** (↑ 3.44%)	0.7003 / **0.7255** (↑ 3.60%)	0.6898 / **0.7073** (↑ 2.54%)	0.7134 / **0.7251** (↑ 1.64%)	0.7190 / **0.7408** (↑ 3.03%)
	ACC	0.7435 / **0.7471** (↑ 0.48%)	0.7414 / **0.7554** (↑ 1.89%)	0.7411 / **0.7545** (↑ 1.81%)	0.7355 / **0.7527** (↑ 2.34%)	0.7400 / **0.7570** (↑ 2.30%)	0.7426 / **0.7626** (↑ 2.69%)
	RMSE	0.4212 / **0.4204** (↓ 0.19%)	0.4191 / **0.4116** (↓ 1.79%)	0.4169 / **0.4086** (↓ 1.99%)	0.4202 / **0.4130** (↓ 1.71%)	0.4155 / **0.4085** (↓ 1.68%)	0.4148 / **0.4073** (↓ 1.81%)
Synthetic (MNAR 0.2)	AUC	0.6697 / **0.6759** (↑ 0.93%)	0.7053 / **0.7232** (↑ 2.54%)	0.7165 / **0.7344** (↑ 2.50%)	0.7029 / **0.7225** (↑ 2.79%)	0.7247 / **0.7377** (↑ 1.79%)	0.7313 / **0.7511** (↑ 2.71%)
	ACC	0.7360 / **0.7384** (↑ 0.33%)	0.7342 / **0.7503** (↑ 2.19%)	0.7341 / **0.7513** (↑ 2.34%)	0.7331 / **0.7504** (↑ 2.36%)	0.7449 / **0.7537** (↑ 1.18%)	0.7495 / **0.7565** (↑ 0.93%)
	RMSE	0.4252 / **0.4245** (↓ 0.16%)	0.4231 / **0.4130** (↓ 2.39%)	0.4195 / **0.4103** (↓ 2.19%)	0.4219 / **0.4131** (↓ 2.09%)	0.4150 / **0.4090** (↓ 1.45%)	0.4132 / **0.4097** (↓ 0.85%)
Synthetic (MNAR 0.0)	AUC	0.6778 / **0.6814** (↑ 0.53%)	0.7016 / **0.7213** (↑ 2.81%)	0.7145 / **0.7326** (↑ 2.53%)	0.7003 / **0.7152** (↑ 2.13%)	0.7245 / **0.7358** (↑ 1.56%)	0.7323 / **0.7476** (↑ 2.09%)
	ACC	0.7351 / **0.7416** (↑ 0.88%)	0.7361 / **0.7509** (↑ 2.01%)	0.7408 / **0.7530** (↑ 1.65%)	0.7337 / **0.7481** (↑ 1.96%)	0.7404 / **0.7536** (↑ 1.78%)	0.7502 / **0.7603** (↑ 1.35%)
	RMSE	0.4230 / **0.4222** (↓ 0.19%)	0.4218 / **0.4131** (↓ 2.06%)	0.4182 / **0.4101** (↓ 1.94%)	0.4210 / **0.4144** (↓ 1.57%)	0.4148 / **0.4093** (↓ 1.33%)	0.4126 / **0.4096** (↓ 0.73%)

Table 3: Performance comparison on synthetic datasets with varying degrees of MNAR. Rows represent datasets (MNAR degree), and columns represent models. The results are presented in the format of "*Original* / ***+TSDR*** (%Improv.)". ↑ indicates a numerical increase, ↓ indicates a numerical decrease. Bold values highlight the better performance in each pair.

Table 4: Ablation study on the effects of joint learning and the placement of temporal smoothness (TS). "TS on Imp." denotes temporal smoothness applied to the imputation branch; "TS on KT" denotes temporal smoothness applied to the KT backbone.

Backbone	DR	Joint Learn.	TS on Imp.	TS on KT	AUC ↑	ACC ↑	RMSE ↓
DKT	✓				0.7258	0.6796	0.4533
	✓	✓			0.7300	0.6792	0.4520
	✓	✓		✓	0.7267	0.6753	0.4545
	✓	✓	✓		**0.7319**	**0.6822**	**0.4510**
AKT	✓				0.7420	0.6932	0.4576
	✓	✓			0.7608	0.7067	0.4443
	✓	✓		✓	0.7596	0.7055	0.4467
	✓	✓	✓		**0.7719**	**0.7136**	**0.4407**
SimpleKT	✓				0.7466	0.6840	0.4560
	✓	✓			0.7635	0.7075	**0.4465**
	✓	✓		✓	0.7599	0.7078	0.4504
	✓	✓	✓		**0.7716**	**0.7110**	0.4466
FoLiBiKT	✓				0.7415	0.6870	0.4638
	✓	✓			0.7646	0.7104	0.4406
	✓	✓		✓	0.7608	0.7083	0.4447
	✓	✓	✓		**0.7703**	**0.7126**	**0.4403**
SparseKT	✓				0.7639	0.7047	0.4477
	✓	✓			0.7709	0.7150	0.4499
	✓	✓		✓	0.7656	0.7090	0.4456
	✓	✓	✓		**0.7775**	**0.7197**	**0.4441**
DisKT	✓				0.7669	0.7121	0.4441
	✓	✓			0.7844	0.7222	0.4365
	✓	✓		✓	0.7818	0.7212	0.4406
	✓	✓	✓		**0.7870**	**0.7262**	**0.4364**

best AUC and ACC on all six backbones. These results verify that the proposed design is backbone-agnostic and can be integrated with both recurrent and attention-based KT models.

Table 5 reports the time and memory overhead introduced by DR. The additional cost is moderate in terms of wall-clock time: the per-epoch training time increases by 0.21s for DKT and by 0.78–1.68s for attention-based backbones. The memory overhead is more noticeable for attention-based models, mainly because the imputation branch introduces additional sequence-level representations and joint optimization requires keeping extra intermediate activations during training. Nevertheless, the additional time and memory overhead is incurred only during training, since the DR phase is used exclusively to guide the KT backbone during optimization.

Table 5: Wall-Clock Time and Memory Overhead.

Backbone	Base t/Epoch (s)	+DR t/Epoch (s)	Δt (s)	Base Mem. (MB)	+DR Mem. (MB)	ΔMem. (MB)
DKT	0.10	0.31	+0.21	1841.2	1901.2	+60.0
AKT	0.43	2.04	+1.61	2037.2	8945.2	+6908.0
SimpleKT	0.23	1.01	+0.78	1815.2	9365.2	+7550.0
FoliBiKT	0.43	2.11	+1.68	2235.2	9383.2	+7148.0
SparseKT	0.25	1.22	+0.97	1837.2	10527.2	+8690.0
DisKT	0.37	1.60	+1.23	1837.2	9119.2	+7282.0

During inference, the DR phase is not involved, and prediction is performed using only the KT backbone without introducing any extra computational overhead. These results indicate that DR improves the prediction accuracy of knowledge tracing with additional cost limited to the training stage.

Appendix E Theoretical Analysis and Main Results

In this section, we establish the theoretical foundation of our framework. We first quantify the generalization error of the Doubly Robust estimator in the sequential setting, decomposing it into bias and variance components. Crucially, we identify that the estimation risk is dominated by the variance of imputation errors. To address this, we provide the theoretical justification for the **Temporal Smoothness** constraint, demonstrating that it effectively tightens the generalization bound by restricting the geometric complexity of the latent trajectory. Detailed proofs for all theorems are provided in Appendix F.

E.1 Problem Setup and Generalization Bound

Let T be the sequence length and $\mathcal{C}$ be the set of skills. The dataset $\mathcal{D}_{\text{full}}$ consists of pairs (t, c) with size N. Let $e_{t+1,c}$ be the true prediction error, $\hat{e}_{t+1,c}$ be the imputed error, and $\hat{p}_{t+1,c}$ be the estimated propensity. We define the **True Risk** $\mathcal{R}_{\text{true}} = \frac{1}{N}\sum e_{t+1,c}$ and the **DR Estimator** $\hat{\mathcal{R}}_{\text{DR}} = \frac{1}{N}\sum(\hat{e}_{t+1,c} + \frac{o_{t+1,c}(e_{t+1,c}-\hat{e}_{t+1,c})}{\hat{p}_{t+1,c}})$.

Our first main result establishes the generalization bound for the DR estimator in this context:

Theorem 1 (Generalization Bound of DR Estimator)**.** *Assuming bounded errors, for any finite hypothesis space $\mathcal{H}$,*

with probability at least $1-\eta$, the true risk is bounded by:

$$\mathcal{R}_{true} \leq \hat{\mathcal{R}}_{DR} + \underbrace{\frac{1}{N}\sum_{(t,c)\in\mathcal{D}_{full}} |\Delta_{t+1,c}\cdot\delta_{t+1,c}|}_{\textit{Bias Term}} + \underbrace{\sqrt{\frac{\ln(2|\mathcal{H}|/\eta)}{2N^2}\sum_{(t,c)\in\mathcal{D}_{full}}\left(\frac{\delta_{t+1,c}}{\hat{p}_{t+1,c}}\right)^2}}_{\textit{Variance Term}}, \tag{1}$$

where $\delta_{t,c} = e_{t,c} - \hat{e}_{t,c}$ is the imputation error deviation, and $\Delta_{t,c}$ is the relative propensity error.

Discussion. Theorem 1 reveals the trade-off in the DR estimator. The **Bias Term** benefits from the double robustness property: it remains small if either the propensity model ($\Delta \approx 0$) or the imputation model ($\delta \approx 0$) is accurate. However, the critical bottleneck is the **Variance Term**, specifically the summation $\sum(\delta_{t,c}/\hat{p}_{t,c})^2$. Since this term scales with the squared imputation error, instability in the imputation model g_ϕ can loosely bound the true risk, leading to poor generalization. Therefore, to stabilize training and tighten this bound, we propose joint training of the prediction and imputation models to simultaneously minimize both prediction and imputation inaccuracies. In particular, reducing the complexity of the imputation model is imperative, as it effectively controls and decreases $\delta_{t,c}$.

E.2 Theoretical Justification of Temporal Smoothness

Generalization Bound via Rademacher Complexity

Let $\mathcal{G}$ be the class of imputation functions parameterized by ϕ. We consider the learning of g_ϕ on the sequence of latent states $H = \{h_1, h_2, \ldots, h_T\}$. Assume $h_t \in \mathbb{R}^d$, and functions in $\mathcal{G}$ satisfy L_g-Lipschitz continuity and boundedness $|g(h)| \leq M$.

Theorem 2 (Generalization Bound for Imputation Model, adapted from [Mohri *et al.*, 2018])**.** *For any $\eta > 0$, with probability at least $1-\eta$, the generalization error of the imputation model is bounded by:*

$$\mathbb{E}[\delta^2] \leq \hat{\mathbb{E}}[\delta^2] + 2\mathfrak{R}_T(\mathcal{G}\circ H) + \sqrt{\frac{\log(1/\eta)}{2T}}, \tag{2}$$

where $\mathfrak{R}_T(\mathcal{G}\circ H)$ denotes the empirical Rademacher Complexity of the function class $\mathcal{G}$ acting on the input trajectory H.

Theorem 2 indicates that minimizing the Rademacher complexity $\mathfrak{R}_T$ is essential for reducing the true error variance.

To minimize the critical variance term identified above, we employ Statistical Learning Theory to link the **Temporal Smoothness** loss, $\mathcal{L}_{smooth} = \sum\|h_t - h_{t-1}\|_2^2$, to the Rademacher complexity of the model.

Theorem 3 (Smoothness-Regularized Complexity Bound)**.** *Assume the imputation function class $\mathcal{G}$ is L_g-Lipschitz continuous. Minimizing the Temporal Smoothness loss $\mathcal{L}_{smooth}$ effectively reduces the Empirical Rademacher Complexity $\mathfrak{R}_T$ of the imputation model. Specifically:*

$$\mathfrak{R}_T(\mathcal{G}\circ H) \leq \frac{12}{\sqrt{T}}\int_0^D \sqrt{\mathcal{C}_{smooth}(\epsilon)\cdot\log\left(\frac{2M}{\epsilon}+1\right)}d\epsilon, \tag{3}$$

where the complexity term $\mathcal{C}_{smooth}(\epsilon) = \frac{L_g\sqrt{T\cdot\mathcal{L}_{smooth}}}{\epsilon} + 1$ is monotonically increasing with respect to $\mathcal{L}_{smooth}$.

Discussion. Theorem 3 provides the theoretical backbone for our proposed regularizer. It establishes a causal chain:

1. Minimizing $\mathcal{L}_{smooth}$ restricts the total path length of the latent trajectory H in the high-dimensional space.
2. A shorter path length reduces the *Covering Number* of the trajectory (geometrically simpler).
3. A reduced Covering Number lowers the *Rademacher Complexity* of the function class $\mathcal{G}$ operating on H.
4. Lower complexity tightens the generalization bound of the imputation error δ^2 via standard Rademacher bounds, thereby directly minimizing the **Variance Term** in Eq. (1).

Appendix F Detailed Proofs and Derivations

This section provides the rigorous mathematical derivations for the theorems presented in Appendix E. We explicitly detail the concentration inequalities and the chaining technique used to bound the model complexity.

F.1 Proof of Theorem 1

The proof relies on decomposing the true risk into bias and variance components and applying concentration inequalities for sequential data.

Bias Analysis

Lemma 1 (Bias of the DR Estimator)**.** *The bias of the DR estimator is bounded by the product of propensity error and imputation error:*

$$\left|\mathcal{R}_{true} - \mathbb{E}_O[\hat{\mathcal{R}}_{DR}]\right| \leq \frac{1}{N}\sum_{(t,c)\in\mathcal{D}_{full}} |\Delta_{t+1,c}\cdot\delta_{t+1,c}|, \tag{4}$$

Proof. Recall that $\mathbb{E}[o_{t,c}] = p_{t,c}$. Taking the expectation of the estimator over observation indicators O:

$$\begin{aligned}\mathbb{E}_O[\hat{\mathcal{R}}_{\text{DR}}] &= \frac{1}{N}\sum_{(t,c)\in\mathcal{D}_{\text{full}}}\left(\hat{e}_{t+1,c} + \frac{p_{t+1,c}}{\hat{p}_{t+1,c}}(e_{t+1,c} - \hat{e}_{t+1,c})\right)\\ &= \frac{1}{N}\sum_{(t,c)\in\mathcal{D}_{\text{full}}}\left(\hat{e}_{t+1,c} + \frac{p_{t,c}}{\hat{p}_{t+1,c}}\delta_{t+1,c}\right).\end{aligned} \tag{5}$$

Substituting the true risk definition $\mathcal{R}_{\text{true}} = \frac{1}{N}\sum e_{t+1,c} =$

$\frac{1}{N}\sum(\hat{e}_{t+1,c}+\delta_{t+1,c})$ into the bias gap:

$$\begin{aligned}
&\text{Bias}\\
&= \left|\frac{1}{N}\sum_{(t,c)}(\hat{e}_{t+1,c}+\delta_{t+1,c}) - \frac{1}{N}\sum_{(t,c)}\left(\hat{e}_{t+1,c}+\frac{p_{t+1,c}}{\hat{p}_{t+1,c}}\delta_{t+1,c}\right)\right|\\
&= \left|\frac{1}{N}\sum_{(t,c)}\delta_{t+1,c}\left(1-\frac{p_{t+1,c}}{\hat{p}_{t+1,c}}\right)\right|\\
&= \left|\frac{1}{N}\sum_{(t,c)}\delta_{t+1,c}\left(\frac{\hat{p}_{t+1,c}-p_{t+1,c}}{\hat{p}_{t+1,c}}\right)\right|.
\end{aligned} \tag{6}$$

Using the definition of relative propensity error $\Delta_{t,c}$ and the triangle inequality $|\sum x| \le \sum |x|$, we obtain the lemma. □

Variance Analysis via Azuma-Hoeffding

Lemma 2 (Concentration for a Single Hypothesis)**.** *For a fixed hypothesis h, with probability at least $1-\eta'$, the deviation is bounded by:*

$$\left|\hat{\mathcal{R}}_{DR} - \mathbb{E}[\hat{\mathcal{R}}_{DR}]\right| \le \sqrt{\frac{\ln(2/\eta')}{2N^2}\sum_{(t,c)\in\mathcal{D}_{full}}\left(\frac{\delta_{t+1,c}}{\hat{p}_{t+1,c}}\right)^2}. \tag{7}$$

Proof. Let $Z_{t,c} = \hat{e}_{t,c} + \frac{o_{t,c}\delta_{t,c}}{\hat{p}_{t,c}}$ be the random variable for each sample. Since student interactions are sequential, the estimation errors conditional on history form a **Martingale Difference Sequence (MDS)**. The range of each term $Z_{t+1,c}$ is determined by the random variable $o_{t+1,c} \in \{0,1\}$:

$$\begin{aligned}
&\text{Range}(Z_{t+1,c})\\
&= \left|\left(\hat{e}_{t+1,c}+\frac{1\cdot\delta_{t+1,c}}{\hat{p}_{t+1,c}}\right) - \left(\hat{e}_{t+1,c}+\frac{0\cdot\delta_{t+1,c}}{\hat{p}_{t+1,c}}\right)\right|\\
&= \left|\frac{\delta_{t+1,c}}{\hat{p}_{t+1,c}}\right|.
\end{aligned} \tag{8}$$

Applying the **Azuma-Hoeffding inequality** [Azuma, 1967] for sum of bounded martingale differences:

$$\begin{aligned}
P\left(|\hat{\mathcal{R}}_{\text{DR}} - \mathbb{E}[\hat{\mathcal{R}}_{\text{DR}}]| \ge \epsilon\right) &\le 2\exp\left(\frac{-2(N\epsilon)^2}{\sum_{(t,c)}(\text{Range}_{t+1,c})^2}\right)\\
&= 2\exp\left(\frac{-2N^2\epsilon^2}{\sum(\delta_{t+1,c}/\hat{p}_{t+1,c})^2}\right).
\end{aligned} \tag{9}$$

Setting the right-hand side to η' and solving for ϵ yields the stated bound. □

Completing Proof of Theorem 1

We then combine the Bias analysis and the Variance concentration to derive the final generalization bound for the hypothesis space $\mathcal{H}$.

Proof. First, we decompose the true risk $\mathcal{R}_{true}$ relative to our DR estimator $\hat{\mathcal{R}}_{\text{DR}}$. By the triangle inequality, the error can be split into a Bias term and a Variance term:

$$\mathcal{R}_{true} \le \hat{\mathcal{R}}_{\text{DR}} + \underbrace{\left|\mathcal{R}_{true} - \mathbb{E}[\hat{\mathcal{R}}_{\text{DR}}]\right|}_{\text{Bias}} + \underbrace{\left|\mathbb{E}[\hat{\mathcal{R}}_{\text{DR}}] - \hat{\mathcal{R}}_{\text{DR}}\right|}_{\text{Variance}}. \tag{10}$$

The **Bias** term has been bounded in the previous theorem. Our remaining task is to bound the **Variance** term uniformly over the entire hypothesis space.

Lemma 2 provides a concentration bound for a *single fixed* hypothesis h. However, since the learning algorithm selects a hypothesis $\hat{h}$ based on the data, we require the bound to hold for the worst-case scenario over all $h \in \mathcal{H}$.

Let $\mathcal{E}(h)$ denote the event that the deviation for hypothesis h exceeds ϵ. According to Lemma 2, for any fixed h, $P(\mathcal{E}(h)) \le \eta'$. We apply the **Union Bound** to control the probability that *any* hypothesis in $\mathcal{H}$ violates the bound:

$$\begin{aligned}
P\left(\max_{h\in\mathcal{H}}\left|\hat{\mathcal{R}}_{\text{DR}}(h) - \mathbb{E}[\hat{\mathcal{R}}_{\text{DR}}(h)]\right| \ge \epsilon\right) &= P\left(\bigcup_{h\in\mathcal{H}}\mathcal{E}(h)\right)\\
&\le \sum_{h\in\mathcal{H}} P(\mathcal{E}(h))\\
&\le \sum_{h\in\mathcal{H}} \eta' = |\mathcal{H}|\cdot\eta'.
\end{aligned} \tag{11}$$

Finally, to ensure the bound holds with high probability $1-\eta$ over the entire space, we set the total failure probability to η:

$$|\mathcal{H}|\cdot\eta' = \eta \implies \eta' = \frac{\eta}{|\mathcal{H}|}. \tag{12}$$

We now substitute this adjusted η' back into the bound for ϵ derived in Lemma 2. The logarithmic term transforms as follows:

$$\ln(2/\eta') = \ln\left(\frac{2}{\eta/|\mathcal{H}|}\right) = \ln\left(\frac{2|\mathcal{H}|}{\eta}\right). \tag{13}$$

Substituting this back into the variance expression:

$$\text{Variance} \le \sqrt{\frac{\ln(2|\mathcal{H}|/\eta)}{2N^2}\sum_{(t,c)\in\mathcal{D}_{\text{full}}}\left(\frac{\delta_{t,c}}{\hat{p}_{t,c}}\right)^2}. \tag{14}$$

Combining this uniform variance bound with the bias bound completes the proof of Theorem 1. □

F.2 Proof of Theorem 3

To prove Theorem 3, we utilize the Chaining technique to derive Dudley's entropy integral bound for the Rademacher complexity [Ma, 2022].

Deriving Dudley's Entropy Integral

Proposition 1. *The empirical Rademacher complexity is bounded by:*

$$\mathfrak{R}_T(\mathcal{G}\circ H) \le \frac{12}{\sqrt{T}}\int_0^D \sqrt{\log\mathcal{N}(\epsilon,\mathcal{G},d_T)}d\epsilon, \tag{15}$$

where D is the diameter of the function class.

Proof. We employ the Chaining Technique [Ma, 2022] to bound the empirical Rademacher complexity. The proof proceeds by constructing a sequence of approximations for the function class $\mathcal{G}$ at increasingly fine scales and bounding the complexity of the differences between these scales.

Construction and Decomposition. Let $\epsilon_0 = D$ be the diameter of the function class. We define a dyadic sequence of scales $\epsilon_j = 2^{-j}\epsilon_0$ for $j \geq 1$. For each scale j, let V_j be a minimal ϵ_j-cover of $\mathcal{G}$ with respect to the empirical metric d_T. For any $g \in \mathcal{G}$, let $\pi_j(g)$ denote the projection of g onto V_j (i.e., the nearest element in V_j). Setting $\pi_0(g) = 0$, and noting that $\pi_j(g)$ converges to g as $j \to \infty$, we can decompose any function g as a telescoping sum:

$$g(h_t) = \sum_{j=1}^{\infty} \left(\pi_j(g)(h_t) - \pi_{j-1}(g)(h_t)\right). \tag{16}$$

By the linearity of the Rademacher complexity and the subadditivity of the supremum, the total complexity is bounded by the sum of complexities at each scale:

$$\begin{aligned}&\mathfrak{R}_T(\mathcal{G} \circ H)\\ &\leq \sum_{j=1}^{\infty} \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{g\in\mathcal{G}} \frac{1}{T}\sum_{t=1}^{T} \sigma_t \underbrace{\left(\pi_j(g)(h_t) - \pi_{j-1}(g)(h_t)\right)}_{\Delta_j(g)(h_t)}\right].\end{aligned} \tag{17}$$

Bounding Increments via Massart's Lemma. We now focus on the difference function $\Delta_j(g) = \pi_j(g) - \pi_{j-1}(g)$. The cardinality of the set of all such differences is at most $|V_j| \cdot |V_{j-1}| \leq |V_j|^2$. To apply Massart's Lemma, we first bound the Euclidean norm of the vector $\boldsymbol{\Delta}_j(g) = (\Delta_j(g)(h_1), \ldots, \Delta_j(g)(h_T))$. Using the definition of the metric d_T and the triangle inequality:

$$\begin{aligned}\|\boldsymbol{\Delta}_j(g)\|_2 &= \sqrt{T} \cdot \|\pi_j(g) - \pi_{j-1}(g)\|_{d_T}\\ &\leq \sqrt{T}\left(\|\pi_j(g) - g\|_{d_T} + \|g - \pi_{j-1}(g)\|_{d_T}\right)\\ &\leq \sqrt{T}(\epsilon_j + \epsilon_{j-1}) = 3\epsilon_j\sqrt{T},\end{aligned} \tag{18}$$

where we used $\epsilon_{j-1} = 2\epsilon_j$. Applying **Massart's Lemma** [Ma, 2022] to the j-th term yields: **Bounding Increments via Massart's Lemma.** We now focus on the difference function class $\mathcal{F}_j = \{\pi_j(g) - \pi_{j-1}(g) : g \in \mathcal{G}\}$. The cardinality of this set is bounded by the product of the covering numbers: $|\mathcal{F}_j| \leq |V_j| \cdot |V_{j-1}| \leq |V_j|^2$.

To apply Massart's Lemma, we identify the two required parameters:

- **Radius (R):** As derived above, the Euclidean norm is bounded by $R = \sup_{f\in\mathcal{F}_j} \|f\|_2 \leq 3\epsilon_j\sqrt{T}$.
- **Cardinality (K):** The number of functions is $K = |\mathcal{F}_j| \leq \mathcal{N}(\epsilon_j)^2$.

Substituting these into the lemma and simplifying the expectation term:

$$\begin{aligned}&\mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_g \frac{1}{T}\sum_{t=1}^{T}\sigma_t\Delta_j(g)(h_t)\right]\\ &= \frac{1}{T}\mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{f\in\mathcal{F}_j}\sum_{t=1}^{T}\sigma_t f(h_t)\right]\\ &\leq \frac{1}{T}\cdot\underbrace{3\epsilon_j\sqrt{T}}_{\text{Radius } R}\cdot\sqrt{2\log(\underbrace{|V_j|^2}_{\text{Card. } K})}\\ &= \frac{3\epsilon_j}{\sqrt{T}}\sqrt{2\cdot 2\log|V_j|}\\ &= \frac{6\epsilon_j}{\sqrt{T}}\sqrt{\log\mathcal{N}(\epsilon_j,\mathcal{G},d_T)}.\end{aligned} \tag{19}$$

Finally, we aggregate these bounds. Using the relation $\epsilon_j = 2(\epsilon_j - \epsilon_{j+1})$, we can interpret the sum as a Riemann approximation. Since the covering number is monotonic, we have $\sqrt{\log\mathcal{N}(\epsilon)} \geq \sqrt{\log\mathcal{N}(\epsilon_j)}$ for $\epsilon \in (\epsilon_{j+1}, \epsilon_j]$. This allows us to upper bound the discrete sum by Dudley's entropy integral:

$$\begin{aligned}\mathfrak{R}_T &\leq \sum_{j=1}^{\infty}\frac{6\epsilon_j}{\sqrt{T}}\sqrt{\log\mathcal{N}(\epsilon_j)}\\ &= \frac{12}{\sqrt{T}}\sum_{j=1}^{\infty}(\epsilon_j - \epsilon_{j+1})\sqrt{\log\mathcal{N}(\epsilon_j)}\\ &\leq \frac{12}{\sqrt{T}}\sum_{j=1}^{\infty}\int_{\epsilon_{j+1}}^{\epsilon_j}\sqrt{\log\mathcal{N}(\epsilon)}\,d\epsilon\\ &= \frac{12}{\sqrt{T}}\int_0^{\epsilon_0}\sqrt{\log\mathcal{N}(\epsilon,\mathcal{G},d_T)}\,d\epsilon.\end{aligned} \tag{20}$$

This concludes the proof. □

Linking Smoothness to Path Length and Covering Number

Lemma 3 (Path Length Bound)**.** *Let the path length be $\ell(H) = \sum_{t=1}^{T}\|h_t - h_{t-1}\|_2$. The covering number of the trajectory satisfies:*

$$\mathcal{N}(\delta, H, \|\cdot\|_2) \leq \frac{\ell(H)}{\delta} + 1 \leq \frac{\sqrt{T\cdot\mathcal{L}_{smooth}}}{\delta} + 1. \tag{21}$$

Proof. The proof consists of two steps: relating path length to smoothness, and covering number to path length.

Algebraic Bound. Let $d_t = \|h_t - h_{t-1}\|_2$. By definition, $\ell(H) = \sum d_t$ and $\mathcal{L}_{smooth} = \sum d_t^2$. Applying the Cauchy-Schwarz inequality to vectors $\mathbf{d}$ and $\mathbf{1}$ of dimension T:

$$\begin{aligned}\ell(H) &= \sum_{t=1}^{T}(d_t\cdot 1)\\ &\leq \sqrt{\sum_{t=1}^{T}d_t^2}\cdot\sqrt{\sum_{t=1}^{T}1^2}\\ &= \sqrt{\mathcal{L}_{smooth}}\cdot\sqrt{T}.\end{aligned} \tag{22}$$

Geometric Bound. Consider the trajectory H as a sequential curve of length $L = \ell(H)$. Construct a covering greedily: start at h_1, and place the next ball center at the first point along the curve at least distance δ away. If k balls are needed, the total length must satisfy $L \geq (k-1)\delta$, implying $k \leq L/\delta + 1$. □

Completing Proof of Theorem 3

Proof. Assuming $g \in \mathcal{G}$ is L_g-Lipschitz and bounded by M, the covering number of the function space is controlled by the covering number of the input trajectory H. Specifically, an ϵ-cover of $\mathcal{G}$ under the L_∞ metric can be constructed from a $(\delta = \epsilon/L_g)$-cover of H. Considering the values in range $[-M, M]$, we have the standard bound:

$$\log \mathcal{N}(\epsilon, \mathcal{G}, d_T) \leq \mathcal{N}\left(\frac{\epsilon}{L_g}, H, \|\cdot\|_2\right) \cdot \log\left(\frac{2M}{\epsilon} + 1\right). \tag{23}$$

Substituting the result from Lemma 3 into this inequality, and then plugging the logarithmic term into the Dudley integral derived, yields the final bound in Theorem 3. Since $\mathcal{L}_{smooth}$ is in the numerator, minimizing it minimizes the complexity bound. □

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